Name

Mathematics Advanced Year 11

Introduction to Functions

Book 1

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Syllabus Content

MAV-11-02 uses functions and relations to model, analyse and solve problems

introduction to functions and relations

- Describe a relation between two sets as an association between the elements of one set and the elements of the other set
- Define a real function f of a real variable x as a relation where each element x of a given set is associated with exactly one element y of the second set
- Recognise that a relation is a rule which can be represented by an algebraic formula, a table of values, a set of ordered pairs (x, y) or a graph
- Use the notation $f(x)$ to identify the unique value of y associated with x when working with functions, and refer to it as the value of f at x
- Refer to x in a function $y = f(x)$ as the independent variable of the function, and refer to y as the dependent variable
- Substitute numeric and algebraic expressions into the formulas of functions
- Apply the vertical line test on the graph of a relation to determine whether it represents a function
- Define the domain of a function f as the set of real numbers on which f is defined
- Define the range of a function f as the set of values of $f(x)$ obtained as x varies over the domain of f
- Refer to a value in the domain of a function at which the function is 0 as a zero of the function
- Recognise that the x-intercepts of the graph of $y = f(x)$ are its zeroes, the solutions of $f(x) = 0$, and that the y-intercept is $f(0)$

Relations

ⓘ Relations

A **relation**, $\mathcal{R}$, from set X to set Y is a collection of ordered pairs (x, y) where the first coordinate comes from X and the second coordinate comes from Y.

In other words, there is a relationship between the two sets.

We can say that $\mathcal{R}$ **'maps'** elements of set X to set Y: $\mathcal{R}: x \mapsto y$.

⚠ In set notation, X refers to the entire set, while x refers to individual elements in X.

🔍 **Identify** Relations

Let $X = \{1,3\}$

$Y = \{p, q, r\}$

✗	✓	✓	✓	✗
$\{(1,p),(4,q)\}$	$\{(1,p),(3,q)\}$	$\{(1,r)\}$	$\{(3,p),(3,r)\}$	$\{(1,p),(q,3)\}$

Using the sets X and Y above, decide if the set of ordered pairs are a relation from X to Y.

a	$\{(1,q),(3,p)\}$	**b**	$\{(1,p),(1,r),(3,q)\}$	**c**	$\{(3,p),(2,q)\}$
d	$\{(p,1)\}$	**e**	$\{(3,r)\}$	**f**	$\{(1,3)\}$

ⓘ Representing Relations

We can represent relations using:

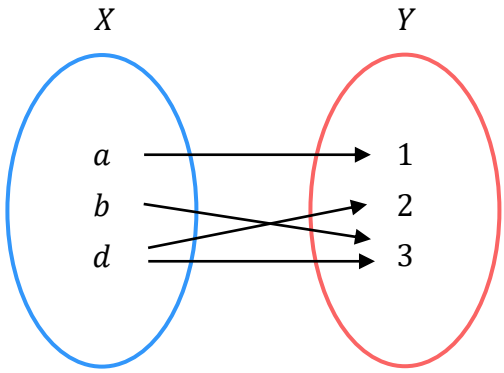
- A set of ordered pairs.
- A table of values.
- An arrow diagram.

🔍 **Interpret** Relation representations

Consider $\mathcal{R} = \{ (a,1), (b,3), (d,2), (d,3) \}$, where $x \mapsto y$.

We can represent this as a table and an arrow diagram:

x	a	b	d	d
y	1	3	2	3



Represent using a table of values and arrow diagram:

a $\{(a,1), (b,2), (d,1), (d,2)\}$

b $\{ (1,5), (1,6), (3,3), (8,4) \}$

c $\{ (7,1), (7,2), (3,1), (2,7), (14,3), (6,2)\}$

ⓘ Real Relations

A relation is **real** when the elements of the two sets are real numbers $\mathbb{R}$.

The connection between the two sets can be arbitrary, or it can be according to some rule.

Relation between students and languages: Not a real relation.

Relation between student heights and masses: A real relation.

Relation between an integer x and $y = x^2$: A real relation.

Real relations are important to this course because they can be graphed on the Cartesian plane, and therefore their behaviour and properties can be easily noticed.

🔍 **Investigation** Real relations

Consider: $\mathcal{R}: \{(x, y) \text{ where } x, y \in \mathbb{R} \text{ and } y = x + 1\}$

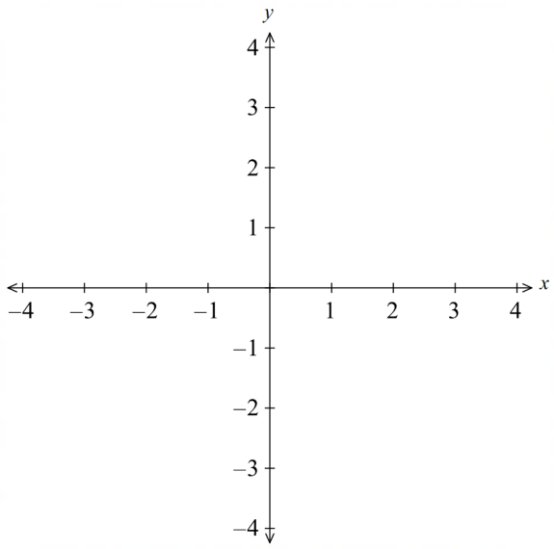
This means:

There is a …… from set …… to set ……, where the elements of both sets are …… numbers, and where the elements are related by the rule ……

Complete the table of values representing this relation:

x	-1	0	1	2	$\frac{1}{3}$	π	p
y							

Sketch the relation on the Cartesian plane.



Since the entire course is concerned with real relations, we can assume that $x, y \in \mathbb{R}$ for all relations from now on and simply state the rule.

$\mathcal{R}: \{(x, y) \text{ where } x, y \in \mathbb{R} \text{ and } y = x + 1\}$ becomes $y = x + 1$

1 $A = \{1,3,5\}, \quad B = \{p, q, r\}$.

For each set of ordered pairs, state whether it is a relation from A to B.
If not, identify the first invalid ordered pair.

a $\{(1,p),(3,q),(5,r)\}$

Yes

b $\{(1,p),(3,q)\}$

Yes

c $\{(1,p),(1,q),(5,r)\}$

Yes

d $\{(1,p),(3,q),(4,r)\}$

No, $(4,r)$

e $\{(1,p),(3,s),(5,r)\}$

No, $(3,s)$

f $\{(1,p),(q,3),(5,r)\}$

No, $(q,3)$

g $\{(5,p)\}$

Yes

2 $A = \{1,2,4\}, \quad B = \{1,3,5,6\}$.

List every ordered pair (a,b) satisfying the rule.

a $b > a$

$\{(1,3),(1,5),(1,6),(2,3),(2,5),(2,6),(4,5),(4,6)\}$

b $b > a + 1$

$\{(1,3),(1,5),(1,6),(2,5),(2,6),(4,6)\}$

c $b > a + 2$

$\{(1,5),(1,6),(2,5),(2,6)\}$

d $b < a$

$\{(2,1),(4,1),(4,3)\}$

e $b \leq a$

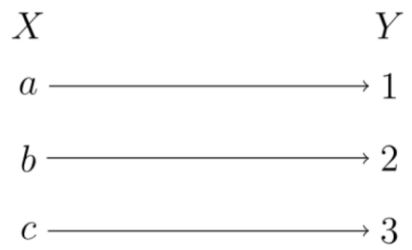
$\{(1,1),(2,1),(4,1),(4,3)\}$

f $b = 2a - 1$

$\{(1,1),(2,3)\}$

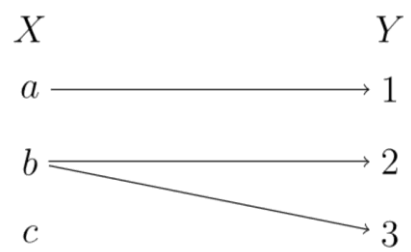
3 Write each relation as a set of ordered pairs.

a



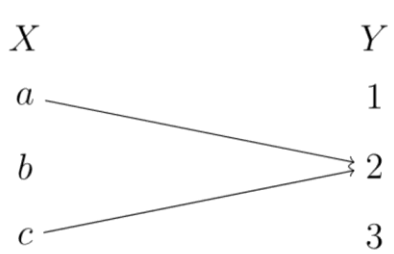
$\{(a,1),(b,2),(c,3)\}$

b



$\{(a,1),(b,2),(b,3)\}$

c

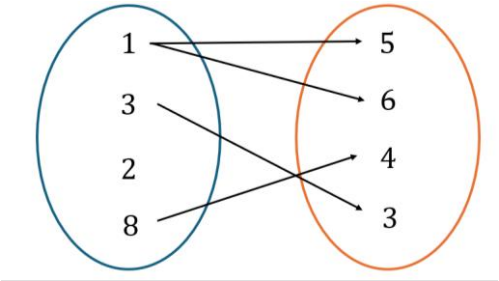


$\{(a,2),(c,2)\}$

Functions

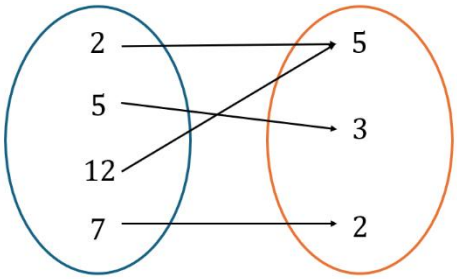
ⓘ Relations and Functions

A **relation** $\mathcal{R}$ connects elements from one set to a second set.



$(1,5), (1,6), (3,3), (8,4)$

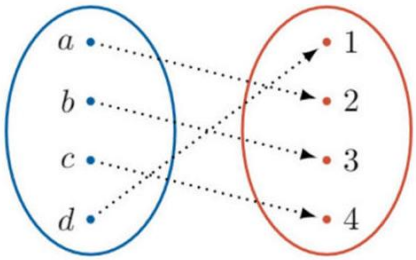
A **function** f is a special relation where each x-value has only one y-value.
(only one arrow leaving each x value)



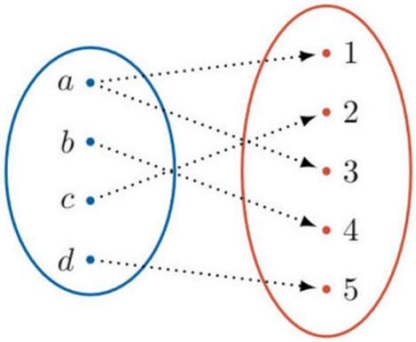
$(2,5), (5,3), (12,5), (7,2)$

🔍 **Identify** whether a relation is a function

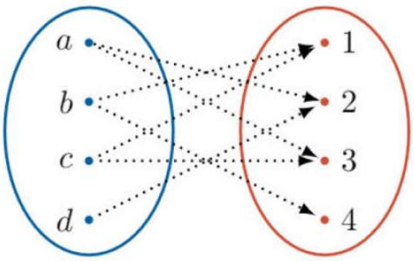
a



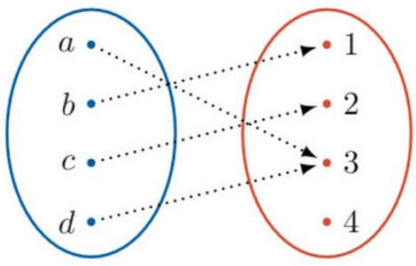
b



c



d

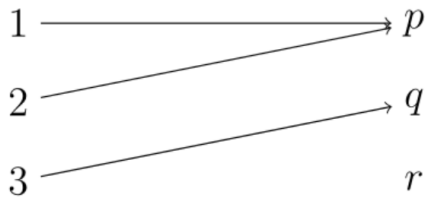


e $\{ (1,2), (4,5), (2,3) \}$

f $\{ (2,5), (2,7), (3,5) \}$

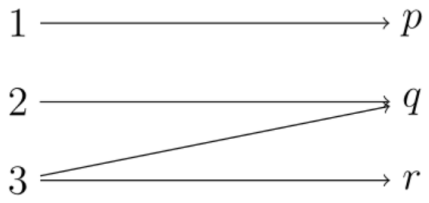
1 Decide whether the following relations are functions.

a



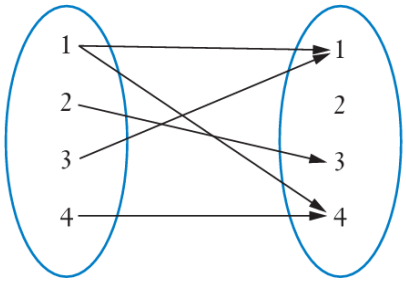
function

b



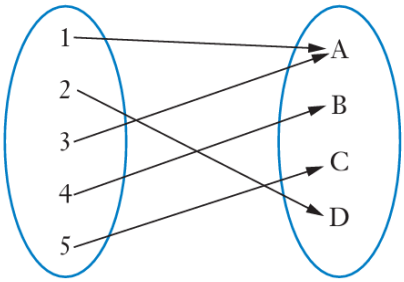
Not function

c



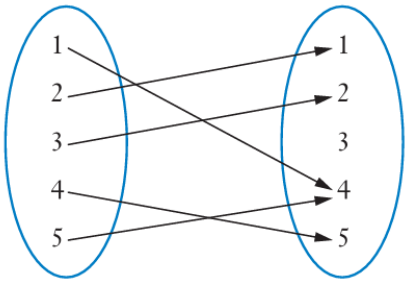
not function

d



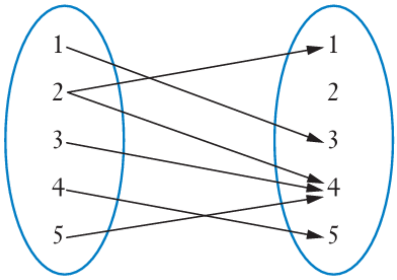
function

e



function

f



not function

g $(-4,1), (3,1), (5,2), (7,-2)$

function

h $(2,2), (3,3), (4,6), (4,9)$

not function

i $(1,1), (2,2), (3,5), (3,7)$

not function

j $(-3,1), (3,1), (8,7), (9,-2), (11,6)$

function

k

x	3	5	8	9	5
y	5	2	-7	3	6

not function

l

x	1	2	5	8	12
y	5	7	± 8	2	1

Not function

m $(x, 5)$ for all real x

function

n $(2, y)$ for all real y

not function

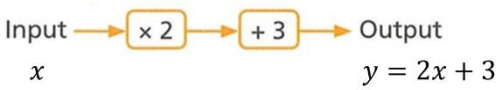
Function Notation

ⓘ Function Machines and Function Notation

Another way to understand functions is to think of it as a 'machine.'

A 'function machine' takes an input number, applies a rule, then gives out an output number.
x is the **input,** y is the **output**.
For a function, every input has at most one output.

An example is the linear function $y = 2x + 3$.



input

$$\underbrace{f(x)}_{\text{output}} = \underbrace{2x + 3}_{\text{function rule}}$$

🔍 **Interpret** Function notation

$f(x) = 2x + 3$ $f(7) = 2(7) + 3$ $= 17$	$f(x) = 2x + 3$ $f(10) = 2(10) + 3$ $= 23$
a $f(x) = 3x - 5$ $f(9) =$	**b** $f(x) = x^2 + 5x$ $f(3) =$
c $g(x) = x^2 + 3$ $g(0) =$	**d** $f(x) = 16^x$ $2f(-1) =$
e $h(x) = \sin x$ $h(30°) =$	**f** $P(x) = x^2 - 2x + 5$ $P(-3) - P(1) =$

Example Substitute an algebraic expression into a function

$f(x) = x^2 + 5x$, find $f(c)$.

$f(c) = c^2 + 5c$

$f(x) = x^2 + 5x$, find $f(x + 1)$.

$f(x + 1) = (x + 1)^2 + 5(x + 1)$

$= x^2 + 2x + 1 + 5x + 5$

$= x^2 + 7x + 6$

Guided Practice

a $f(x) = x^3 + 3$, find $f(n)$.

$n^3 + 3$

b $g(x) = 2x + 3$, find $g(3 - x)$.

$9 - 2x$

c $f(x) = -x^2$, find $f(x + 1)$.

$-x^2 - 2x - 1$

d $f(x) = 3x^2 + 2x$, find $f(x + h)$

$3x^2 + 6xh + 3h^2 + 2x + 2h$

Example Solve $f(x) = c$

$f(x) = 2x - 5$, find x when $f(x) = 13$

$13 = 2x - 5$

$18 = 2x$

$9 = x$

$f(x) = x^2 - 3x$, find x when $f(x) = 10$

$10 = x^2 - 3x$

$0 = x^2 - 3x - 10$

$0 = (x - 5)(x + 2)$

$x = 5, -2$

Guided Practice

a $f(x) = 4x + 11$, find x when $f(x) = 24$

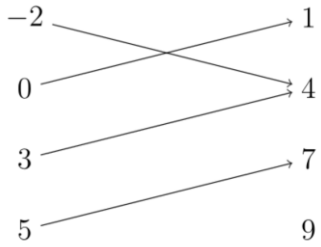
$x = \frac{13}{4}$

b $g(x) = x^2 + 3$, find x when $g(x) = 28$

$x = \pm 5$

1 The following shows inputs x and the function outputs $f(x)$.

x	-2	0	3	5
$f(x)$	4	1	4	7



Write:

a $f(-2)$ — 4

b $f(0)$ — 1

c $f(5)$ — 7

2 $f(x) = x^2 - 2x - 3$. Find:

a $f(0)$ — −3

b $f(3)$ — 0

c $f(-2)$ — 5

3 $f(x) = x^3 + 8$. Find:

a $f(2)$ — 16

b $f(0)$ — 8

c $f(-2)$ — 0

4 $f(x) = 2^x$. Find:

a $f(2)$ — 4

b $f(0)$ — 1

c $f(-2)$ — 0.25

5 $h(x) = 3x - x^2$. Find:

a $h(-3)$ — −18

b $h(1)$ — 2

c $h(5)$ — −10

6 $h(x) = \sqrt{x+4}$. Find:

a $h(-3)$ — 1

b $h(1)$ — $\sqrt{5}$

c $h(5)$ — 3

7

a $f(x) = 3x + 2$.
Find a if $f(a) = 14$.

$a = 4$

b $f(x) = 3x + 2$.
Find a if $f(a) = -4$.

$a = -2$

c $f(x) = x^2 + 1$.
Find all a if $f(a) = 10$.

$a = \pm 3$

d $f(x) = x^2 + 1$.
Find all a if $f(a) = 1$.

$a = 0$

e $f(x) = x^2 + 1$.
Find all real a if $f(a) = 0$.

No real solution

f $f(x) = (x - 2)^2$.
Find all a if $f(a) = 9$.

$a = 5 \text{ or } -1$

g $f(x) = x^2 + x + 1$.
Find all a if $f(a) = 7$.

$a = -3 \text{ or } 2$

h $f(x) = 2x^2 - 8$.
Find all a if $f(a) = 10$.

$a = \pm 3$

8 Find all real a such that $f(a) = 5$.

a $f(x) = x + 1$

$a = 4$

b $f(x) = 2x + 1$

$a = 2$

c $f(x) = x^2 + 1$

$a = \pm 2$

d $f(x) = x^2 + 5$

$a = 0$

e $f(x) = x^2 + 6$

No real solution

9 $f(x) = 3x^2 + 1$. Find the following:

a $f(-3)$

28

b $f(-a)$

$3a^2 + 1$

c $f(a^2)$

$3a^4 + 1$

d $f(2a)$

$12a^2 + 1$

e $f(\sqrt{a})$

$3a + 1$

f $f(ab)$

$3a^2b^2 + 1$

g $f(2a - 5)$

$12a^2 - 60a + 76$

h $f(1 + a)$

$3a^2 + 6a + 4$

i $f\left(\frac{a}{b}\right)$

$\frac{3a^2}{b^2} + 1$

10 For the function $f(x) = 3x + 1$, find:

a $f(1) - 2$

2

b $3f(-1)$

-6

c $f(9) \div f(2)$

4

11 For the function $f(x) = x^2 - 3x + 5$, find:

a $\frac{f(2)+f(3)}{2}$

4

b $\frac{f(0)+4f(2)+f(4)}{6}$

$4\frac{1}{3}$

12 For the function $f(x) = x^2 - 2x - 4$, find:

a $f(1 + \sqrt{5})$

0

b $f(\sqrt{3} - 1)$

$2 - 4\sqrt{3}$

13 Rewrite these functions with function notation $f(x)$.

a $y = 8x$

$f(x) = 8x$

b $y = 9 - x^2$

$f(x) = 9 - x^2$

c $y + 7 = x$

$f(x) = x - 7$

14 Make y the subject then rewrite these functions with function notation.

a $2y + 5 = x$

$f(x) = \frac{x - 5}{2}$

b $3x + 4y + 5 = 0$

$f(x) = -\frac{3x + 5}{4}$

c $x = y^3$

$f(x) = \sqrt[3]{x}$

15 Make y the subject then rewrite these functions with function notation.

a $2x + y = 7$; find $f(3)$.

b $2x + y = 7$; find $f(-3)$.

c $2x + 2y = 8$; find $f(3)$.

d $3x + 2y = 8$; find $f(2)$.

e $3x - 2y = 8$; find $f(2)$.

f $5x - 2y = 1$; find $f(-1)$.

16 For these real functions, explain why the function value cannot be found.

a $F(0)$, when $F(x) = \sqrt{x - 4}$

$F(0) = \sqrt{-4}$, not real

b $g(3)$, when $g(x) = \sqrt{1 - x^2}$

$g(3) = \sqrt{-8}$, not real

c $h(-2)$, when $h(x) = \frac{1}{2+x}$

$h(-2) = \frac{1}{0}$, undefined

d $f(0)$, when $f(x) = \frac{1}{3x}$

$f(0) = \frac{1}{0}$, undefined

17 Find $g(a+1)$ for:

a $g(x) = 2 - x$

$1 - a$

b $g(x) = x^2$

$a^2 + 2a + 1$

c $g(x) = \frac{1}{x-1}$

$\frac{1}{a}$

18 Find $f(t) - 2$ and $f(t-2)$ for:

a $f(x) = 5x + 2$

$5t \mid 5t - 8$

b $f(x) = \sqrt{x}$

$\sqrt{t} - 2 \mid \sqrt{t-2}$

c $f(x) = 2 - x^2$

$-t^2 \mid -t^2 + 4t - 2$

19 Let $g(x) = 4x + 1$

Find and simplify each pair.

a $g(2a)$ and $2g(a)$

$8a + 1 \mid 8a + 2$

b $g(3a)$ and $3g(a)$

$12a + 1 \mid 12a + 3$

c $g(a^2)$ and $\big(g(a)\big)^2$

$4a^2 + 1 \mid 16a^2 + 8a + 1$

d $g(a+1)$ and $g(a) + 1$

$4a + 5 \mid 4a + 2$

e $g(-a)$ and $-g(a)$

$1 - 4a \mid -4a - 1$

20 $f(x) = x^2 + 5x$. Find, in simplest form:

a $\dfrac{f(1+h) - f(1)}{h}$

$7 + h$

b $\dfrac{f(p) - f(q)}{p - q}$

$p + q + 5$

c $\dfrac{f(x+h) - f(x)}{h}$

$2x + h + 5$

21 For each claim, simplify both sides. State **true for all valid** x or **not an identity**.

a If $f(x) = x^2 - x$,
then $f(1-x) = f(x)$.

True for all x

b If $f(x) = x^2$,
then $f(-x) = f(x)$.

True for all x

c If $f(x) = x^2 + 1$,
then $f(-x) = f(x)$.

True for all x

d If $f(x) = x^2 + x$,
then $f(-x) = f(x)$.

Not an identity (only true when x=0)

e If $f(x) = 3x - 2$,
then $f(x+1) = f(x) + 3$.

True for all x

f If $f(x) = 3x - 2$,
then $f(2x) = 2f(x)$.

Not an identity

g If $f(x) = \frac{1}{x}$,
then $f\left(\frac{1}{x}\right) = x$ for $x \neq 0$.

True for all $x \neq 0$

h If $f(x) = x + \frac{1}{x}$,
then $f\left(\frac{1}{x}\right) = f(x)$ for $x \neq 0$.

True for all $x \neq 0$

22 By finding expressions for the LHS and RHS, show that these are true for $f(x) = x^2$.

a $f(4x) = 16f(x)$

b $f(xy) = f(x)f(y)$

c $f\left(\frac{x}{y}\right) = \frac{f(x)}{f(y)}$

d $f(x^2) = \big(f(x)\big)^2$

e $f(x+y) + f(x-y) = 2\big(f(x) + f(y)\big)$

23 Find $f(k) - f(k-1)$ in simplest form for:

a $f(x) = 3x^2 + 2x$

$6k - 1$

b $f(x) = \frac{1}{x}$

$\frac{1}{k(1-k)}$

c $f(x) = 2^x$

2^{k-1}

24 By finding expressions for the LHS and RHS, show that these are true for $f(x) = 4^x$.

a $4f(x) = f(x+1)$

True

b $f(x+1) - f(x) = 3f(x)$

True

25

a $f(x) = 2^x$

Show that $f(x) \times f(-x) = 1$.

b $f(x) = x - x^3$

Show that $f(-x) = -f(x)$.

c $f(x) = \frac{x}{x+1}$

Show that $f(x) + f\left(\frac{1}{x}\right) = 1$.

d $f(x) = \frac{x}{x^2+1}$

Show that $f\left(\frac{1}{x}\right) = f(x)$.

e $f(x) = px^2 - x$.

Find p if $f(-3) = 2f(2)$.

$p = -7$

f $f(x) = \frac{x}{x^2-1}$

Show that $f(x) + f\left(\frac{1}{x}\right) = 0$.

g $f(x) = \frac{a-x}{1+bx}$.

Show that $f\left(\frac{a-x}{1+bx}\right) = x$

h $c(x) = \frac{3^x+3^{-x}}{2}, s(x) = \frac{3^x-3^{-x}}{2}$

Show that $\left(c(x)\right)^2 - \left(s(x)\right)^2 = 1$

Vertical Line Test

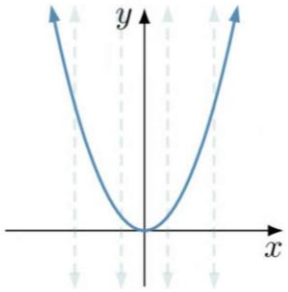
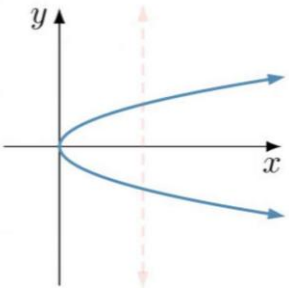
ⓘ The Vertical Line Test

We can plot relations as coordinates (x, y) on a Cartesian plane and make a graph.

For a function, every input has at most one output.

This means any **vertical line** will intersect the graph of a function **at most once**.

If a vertical line intersects a graph **more than once**, then the graph is **not a function**.

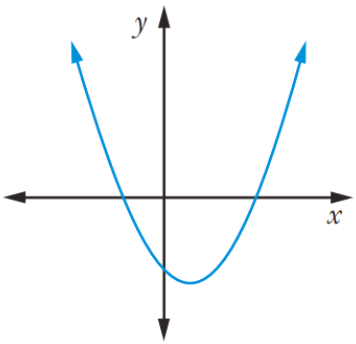
Function	Not a Function
	

A closed circle ● means the point is on the graph.

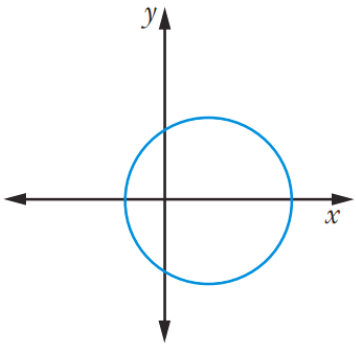
An open circle ○ means it is not.

🔍 **Decide** whether a graph represents a function using the vertical line test

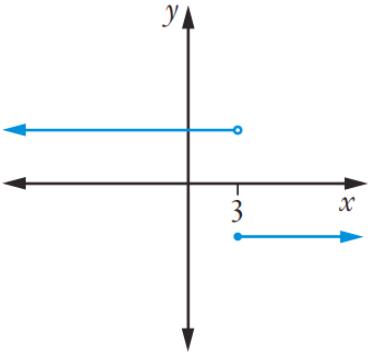
a



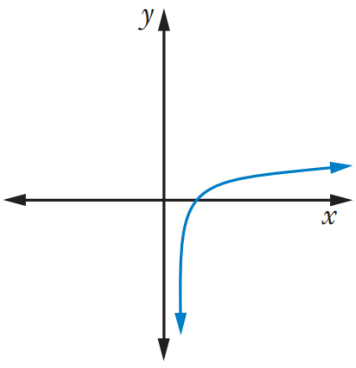
b



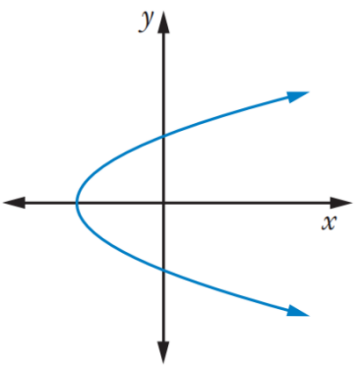
c



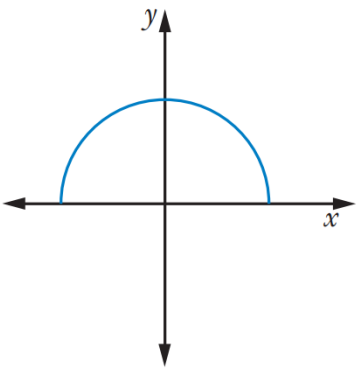
d



e

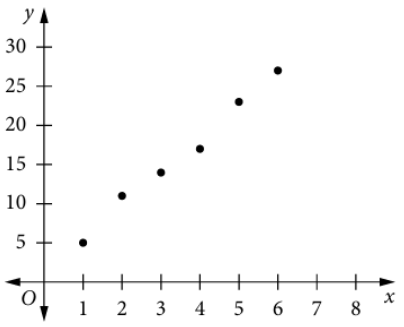


f



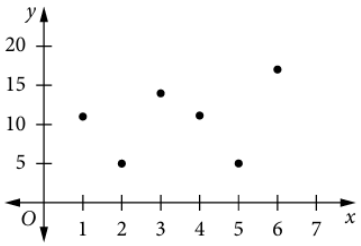
1 Decide whether these relations show a function using the vertical line test.

a



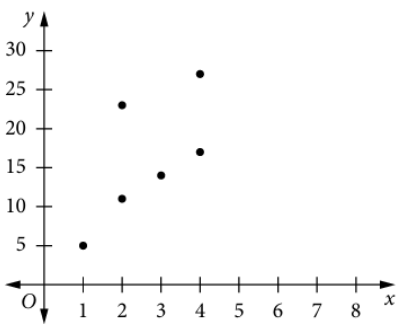
function

b



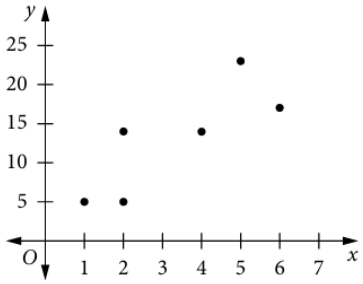
function

c



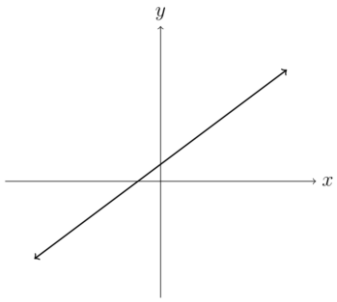
not function

d



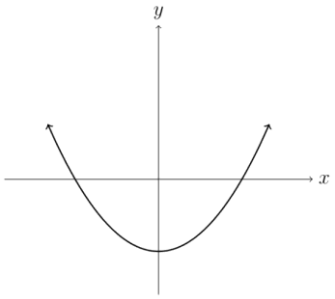
not function

e



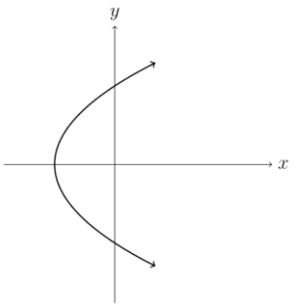
function

f



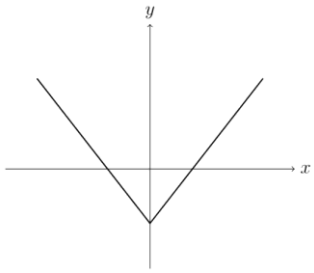
function

g



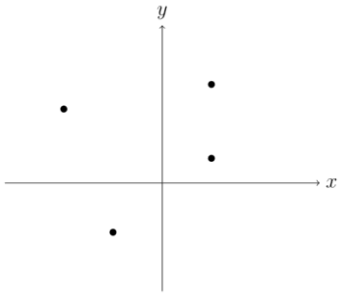
not function

h



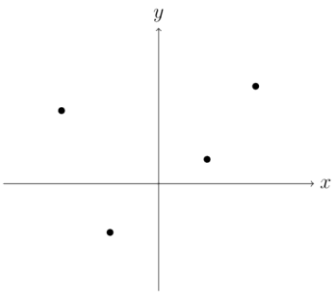
function

i



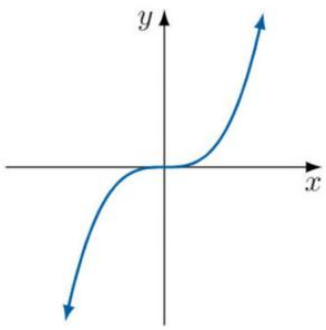
not function

j



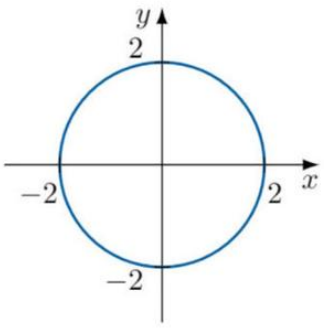
function

k



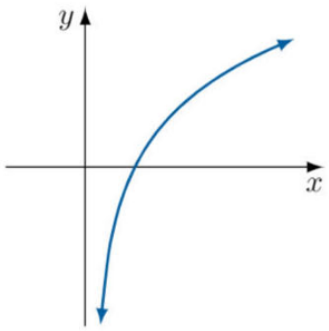
function

l



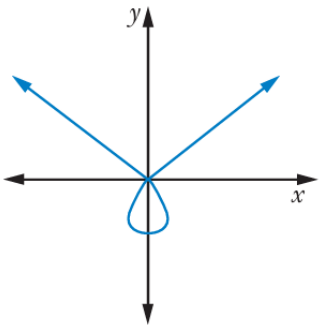
not function

m



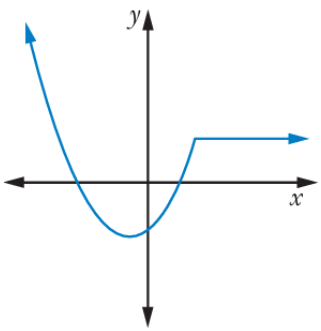
function

n



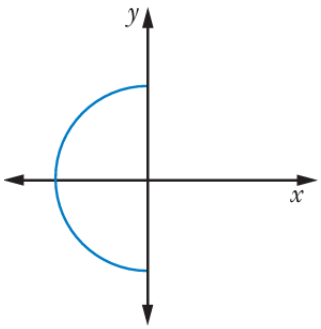
not function

o



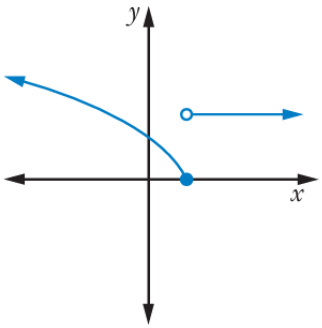
function

p



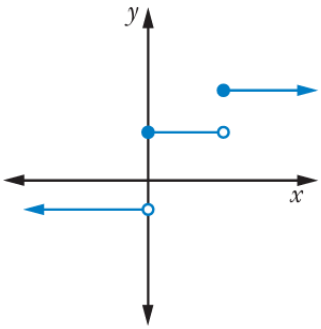
not function

q



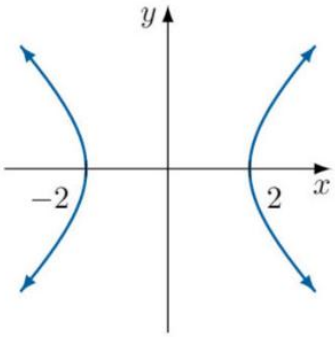
function

r



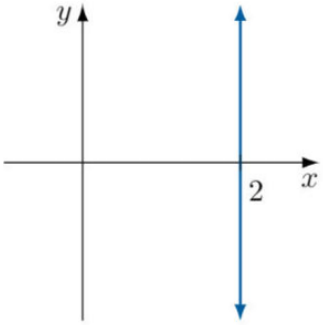
function

s



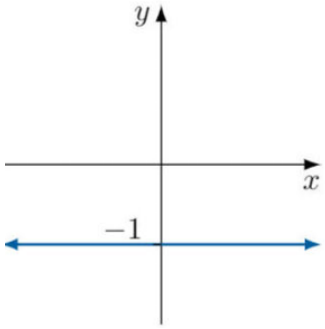
not function

t



not function

u



function

Function Notation and Graphs

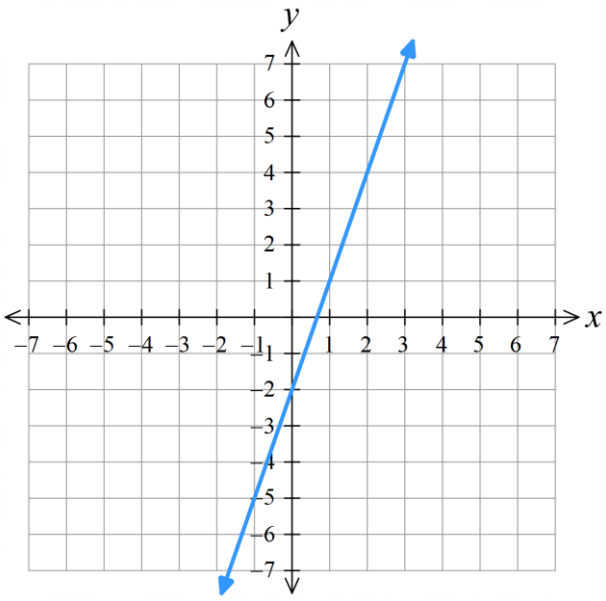
ⓘ $f(a)$ and $f(x) = c$

For a function $y = f(x)$

$f(a)$ means find the y coordinate when $x = a$.

$f(x) = c$ means find the x coordinate when $y = c$.

Interpret Function notation and the graph



Evaluate $f(0)$

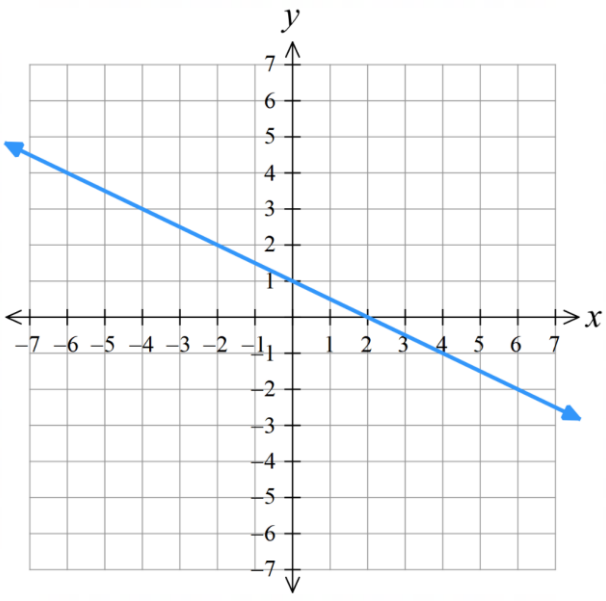
Evaluate $f(1)$

Evaluate $f(-1)$

Evaluate $f(3)$

Find x when $f(x) = 4$

Find x when $f(x) = -2$



Evaluate $f(0)$

Evaluate $f(1)$

Evaluate $f(2)$

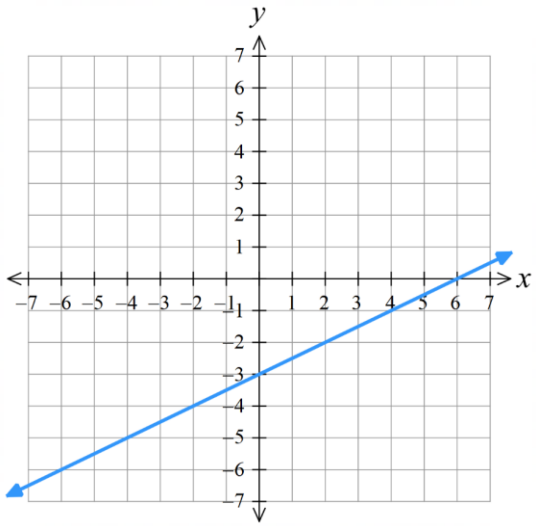
Evaluate $f(-3)$

Find x when $f(x) = 4$

Find x when $f(x) = -2$

1

a



Evaluate $f(0)$

$f(0) = -3$

Evaluate $f(4)$

$f(4) = -1$

Evaluate $f(-5)$

$f(-5) = -5.5$

Find x when $f(x) = -5$

$x = -4$

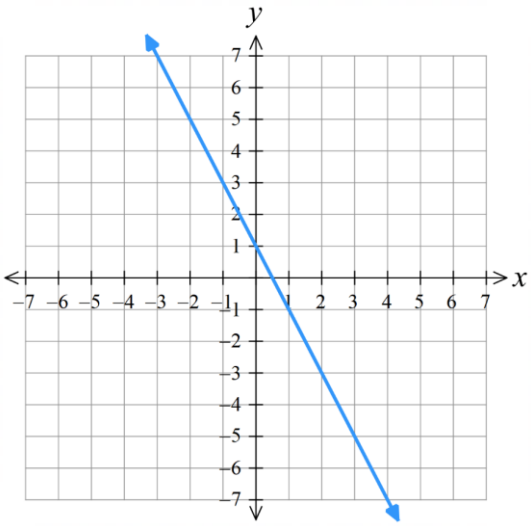
Find x when $f(x) = -1.5$

$x = 3$

Find x when $f(x) = 0$

$x = 6$

b



Evaluate $f(-1)$

$f(-1) = 3$

Evaluate $f(2.5)$

$f(2.5) = -4$

Find x when $f(x) = 2$

$x = -0.5$

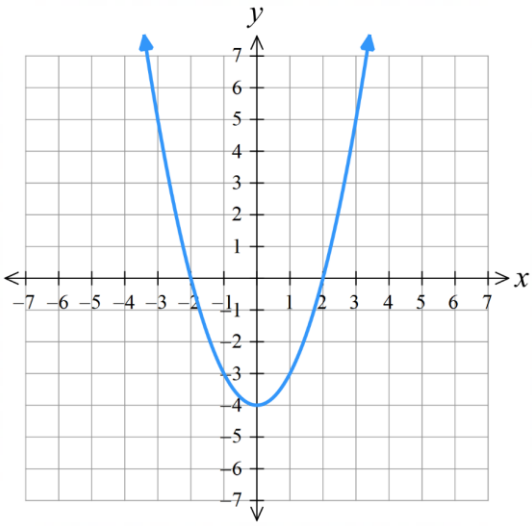
Find x when $f(x) = -7$

$x = 4$

Evaluate $f\left(-\frac{3}{2}\right)$

$f\left(-\frac{3}{2}\right) = 4$

c



Evaluate $f(0)$

$f(0) = -4$

Find x when $f(x) = 0$

$x = -2$ or 2

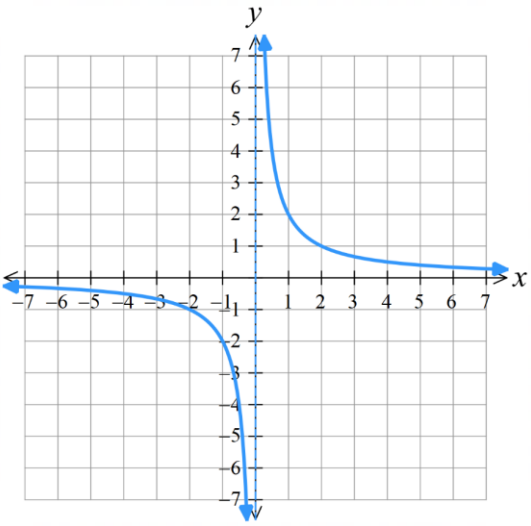
Evaluate $f(1)$

$f(1) = -3$

Find x when $f(x) = 5$

$x = -3$ or 3

d



Evaluate $f(1)$

$f(1) = 2$

Evaluate $f(-2)$

$f(-2) = -1$

Evaluate $f(0)$

$f(0)$ is undefined

Find x when $f(x) = -2$

$x = -1$

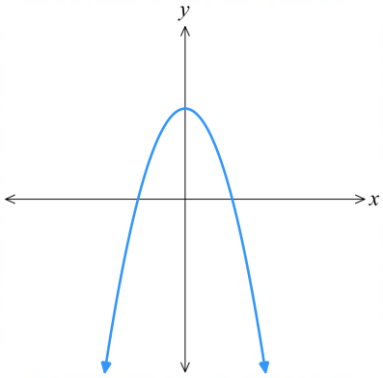
2 The y-intercept of a function is at $x = 0$. Write the y-intercept using function notation.

$f(0)$

3 The x-intercept(s) of a function is at $y = 0$. Write the x-intercept(s) using function notation.

$f(x) = 0$

4 The graph of $f(x) = 4 - x^2$ is shown.



a Find $f(0)$ and show the point on the graph.

(0,4)

b Find $f(x) = 0$ and show the points on the graph.

(–2,0) and (2,0)

c Find $f(4)$ and show the point on the graph.

(4,–12)

5 Consider the following tables of values showing x and $f(x)$.

a

x	-2	-1	0	1	2
$f(x)$	1	3	5	7	9

Evaluate $f(0)$

$f(0) = 5$

Evaluate $f(2)$

$f(2) = 9$

Find x when $f(x) = 1$

$x = -2$

Find x when $f(x) = 3$

$x = -1$

Find x when $f(x) = 9$

$x = 2$

b

x	0	-3	-2	2	1
$f(x)$	-3	6	1	1	-2

Evaluate $f(-2)$

$f(-2) = 1$

Evaluate $f(1)$

$f(1) = -2$

Find x when $f(x) = -2$

$x = 1$

Find x when $f(x) = 1$

$x = -2$ or 2

Evaluate $f(-3)$

$f(-3) = 6$

6 Consider these representations of functions.

a $(-4,1), (3,1), (5,2), (7,-2)$

Evaluate $f(3)$

$f(3) = 1$

Evaluate $f(7)$

$f(7) = -2$

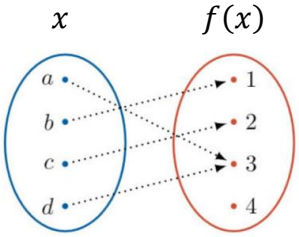
Find x when $f(x) = 2$

$x = 5$

Find x when $f(x) = 1$

$x = -4$ or 3

b



Evaluate $f(a)$

$f(a) = 3$

Find x when $f(x) = 1$

$x = b$

Find x when $f(x) = 4$

No solution

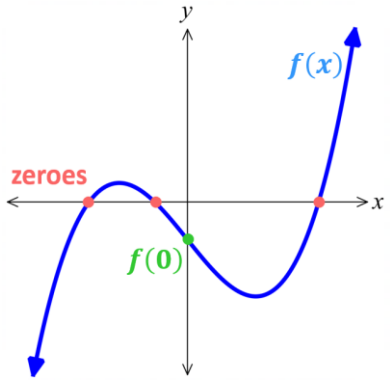
Find x when $f(x) = 3$

$x = a$ or d

Zeroes and $f(0)$

ⓘ Zeroes of a function and $f(0)$

For $y = f(x)$:
$f(0)$ is the **y-intercept**.
A **zero** of the function is the x value where $f(x) = 0$. (i.e., **the x-intercept**).
A function can have many zeroes but only one $f(0)$.

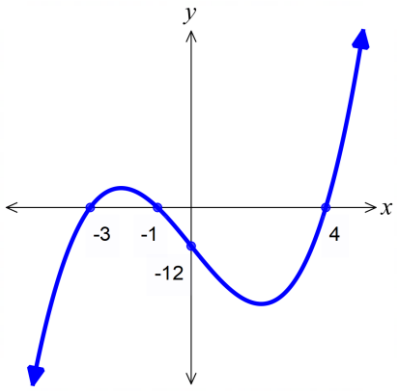


Identify Zeroes of a function and $f(0)$

a

Zeroes:

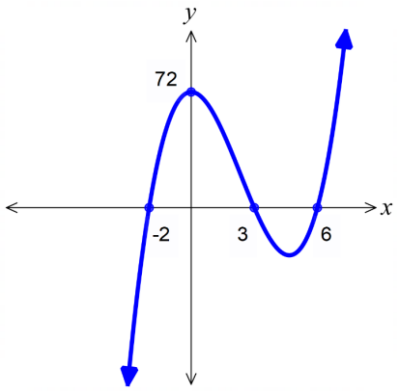
$f(0) =$



b

Zeroes:

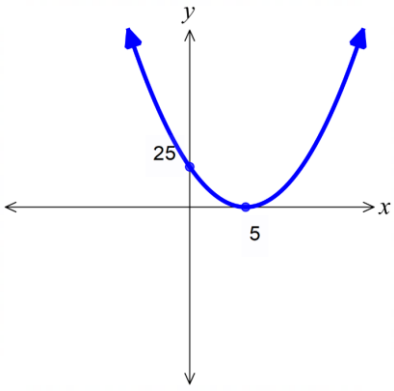
$f(0) =$



c

Zeroes:

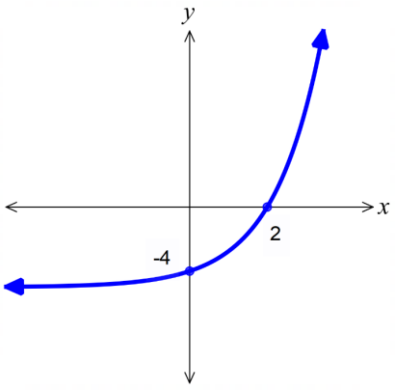
$f(0) =$



d

Zeroes:

$f(0) =$



Key Ideas

1. When you are asked to find $f(0)$, that means find the …… value when …… = 0. In other words, find the ……-intercept.
2. When you are asked to find the zero(es) of a function, that means find where …… = 0, in other words, find the ……-intercept(s).

Interval Notation

Investigation Interval notation

There are two ways of writing the same interval:

$$-1 \leq x < 3$$

$$[-1, 3)$$

1 What is true about the [and) symbols?

- ☐ [and] are used to include a value, just like ≤ and ≥
- ☐ (and) are used to exclude a value, just like < and >
- ☐ Both above statements are true.

2 What does $[2,4]$ mean?

- ○ $2 \leq x \leq 4$
- ○ $2 < x < 4$
- ○ $2 \leq x < 4$

3 What does $[2,4)$ mean?

- ○ $2 \leq x \leq 4$
- ○ $2 < x < 4$
- ○ $2 \leq x < 4$

4 How would you represent $x \geq 0$ in interval notation?

- ○ $[0, \infty)$ since ∞ can't be an included value.
- ○ $[0, \infty]$ since ∞ can be an included value.

5 Write these inequalities using interval notation.

a



b



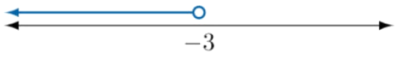
c



d



e



f

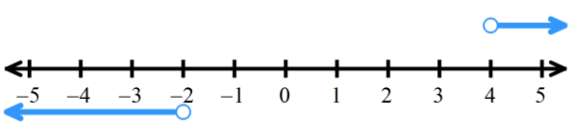


When you have two separate inequalities, we use the union symbol ∪

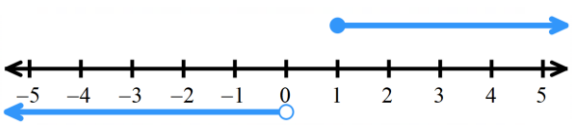
$$x \leq 1, \qquad x \geq 3$$

$$(-\infty, 1] \cup [3, \infty)$$

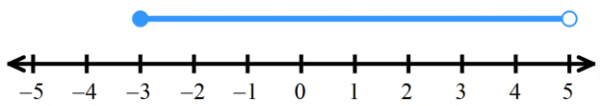
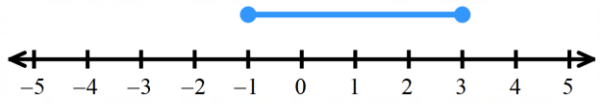
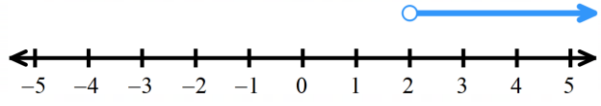
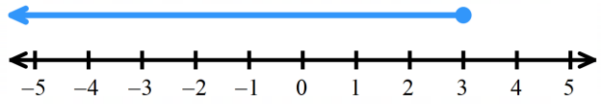
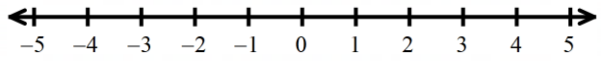
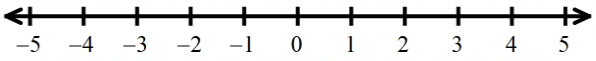
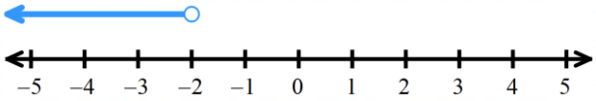
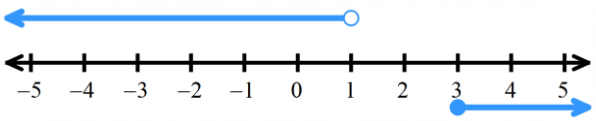
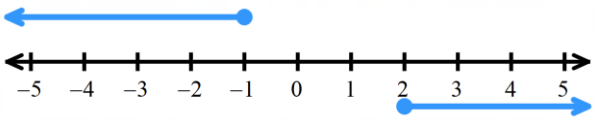
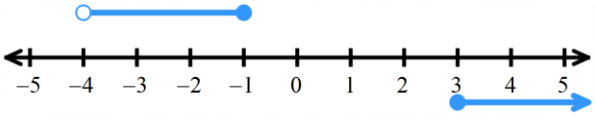
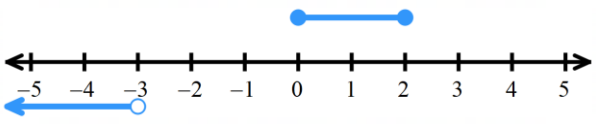
a



b



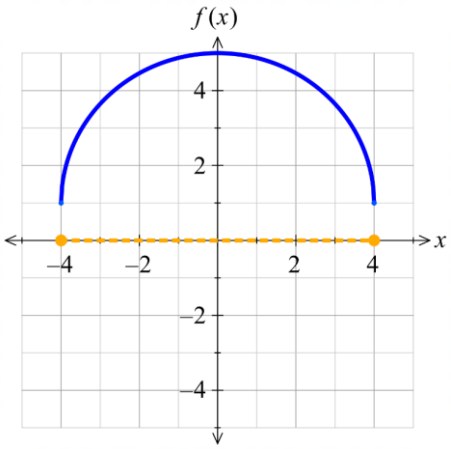
1 Complete the table.

	Inequality	Interval Notation	Graph
a	$-3 \leq x < 5$	$[-3, 5)$	
b	$-1 \leq x \leq 3$	$[-1,3]$	
c	$x > 2$	$(2, \infty)$	
d	$x \leq 3$	$(-\infty, 3]$	
e	$x < 4$	$(-\infty, 4)$	
f	$x \geq 5$	$[5, \infty)$	
g	$x < -2$	$(-\infty, -2)$	
h	$x < 1$ or $x \geq 3$	$(-\infty, 1) \cup [3, \infty)$	
i	$x \leq -1$ or $x \geq 2$	$(-\infty, -1] \cup [2, \infty)$	
j	$-4 < x \leq -1$ or $x \geq 3$	$(-4, -1] \cup [3, \infty]$	
k	$x < -3$ or $0 \leq x \leq 2$	$(-\infty, -3) \cup [0,2]$	

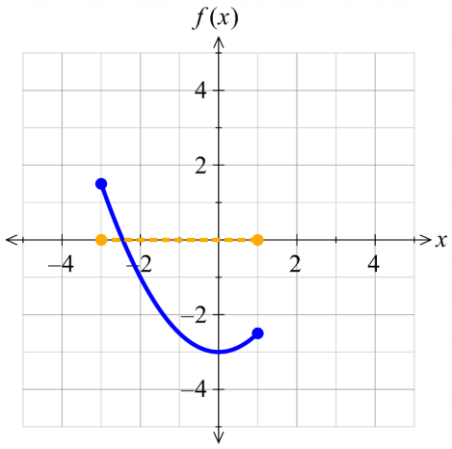
Domain and Range

Investigation Domain

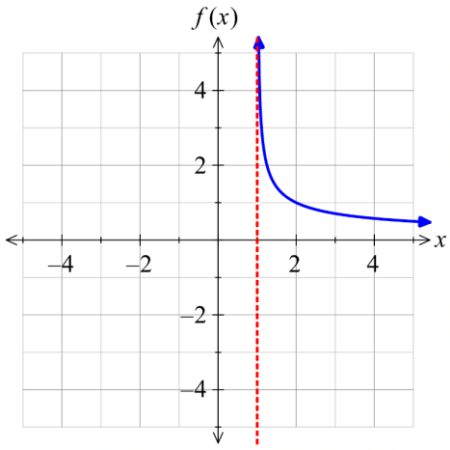
The domain is the **allowable inputs** of a function, or the x values for which a function exists. We can specify a domain with an interval, and we can visualise it as a shadow on the x-axis.



The domain is $-4 \leq x \leq 4$



The domain is …… $\leq x \leq$ ……

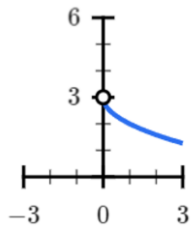


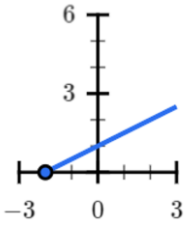
The domain is: (choose one)

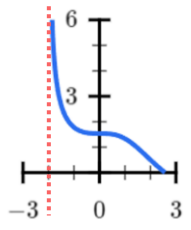
- ☐ $x \geq 1$
- ☐ $x > 1$
- ☐ x can be any real number

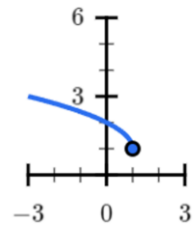
Match the domain to the function.

$x > -2$ $x \geq -2$ $x \leq 1$ $x > 0$



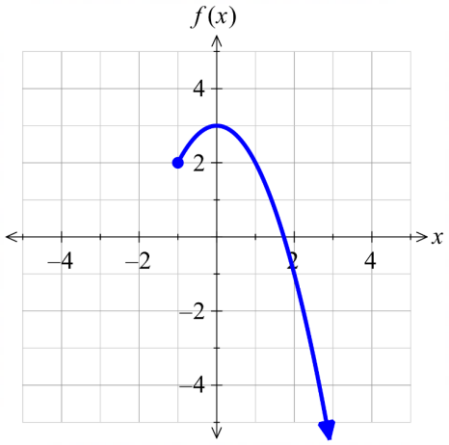


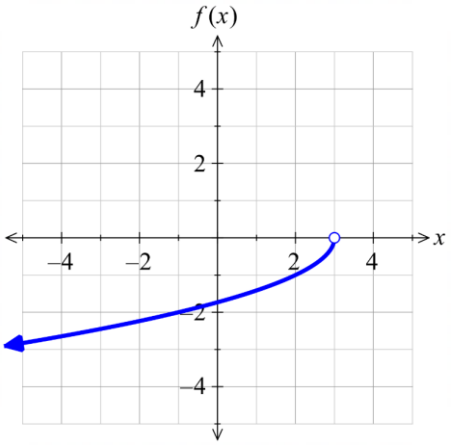




Investigation Domain

What is the domain, using **interval notation**?

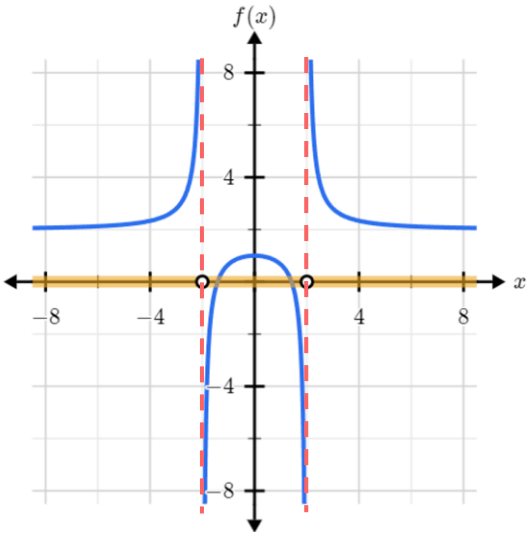




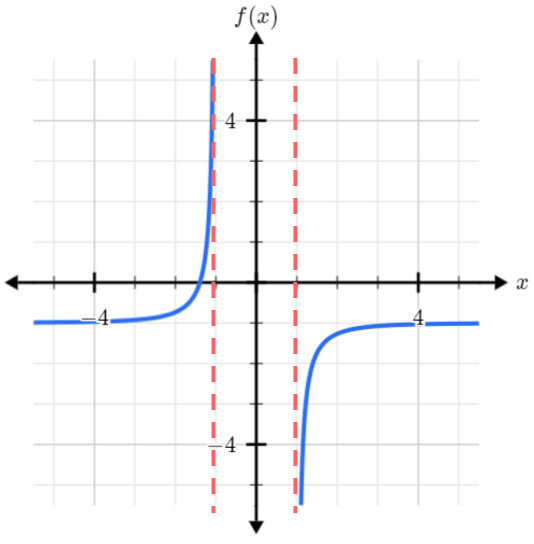
We can combine separate intervals of a domain with the union symbol ∪.

What is the domain of the graphed function?

- $(-\infty, -2) \cup (-2,2) \cup (2, \infty)$
- $(-\infty, -2] \cup [-2,2] \cup [2, \infty)$
- $[-\infty, -2] \cup (-2,2) \cup (2, \infty]$



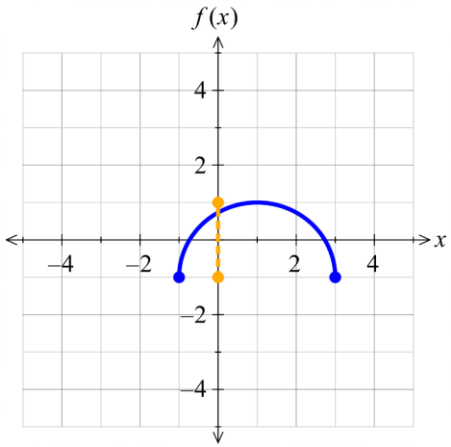
What is the domain of the graphed function?



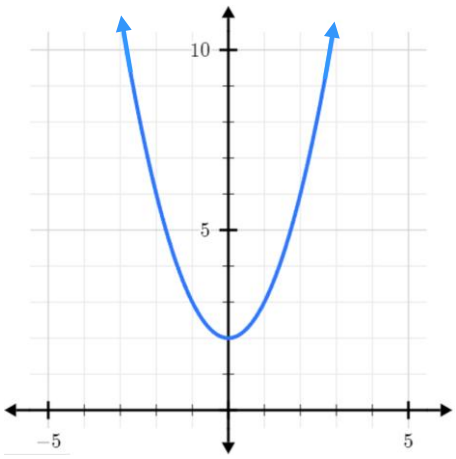
Investigation Range

The range is the possible outputs of a function.

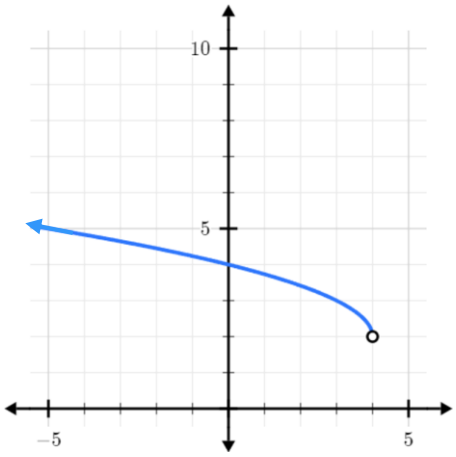
We can specify a range with an interval, and we can visualise it as a shadow on the y-axis.



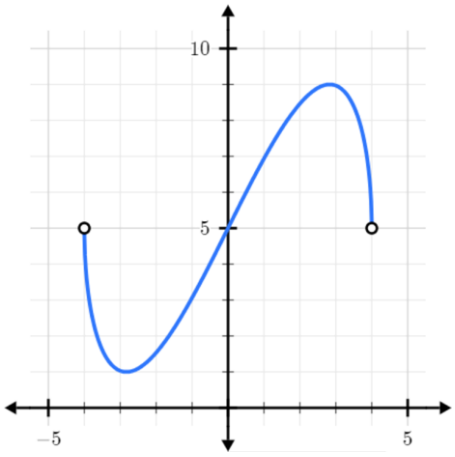
The range is $[-1,1]$



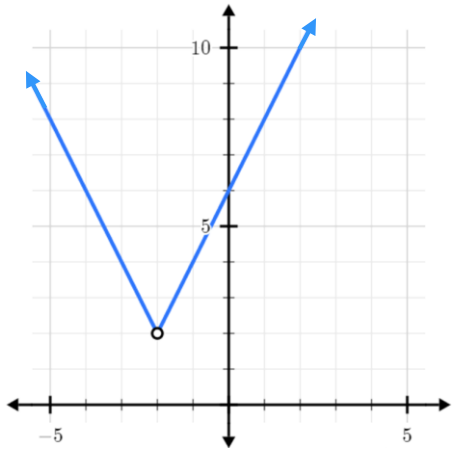
The range is ……



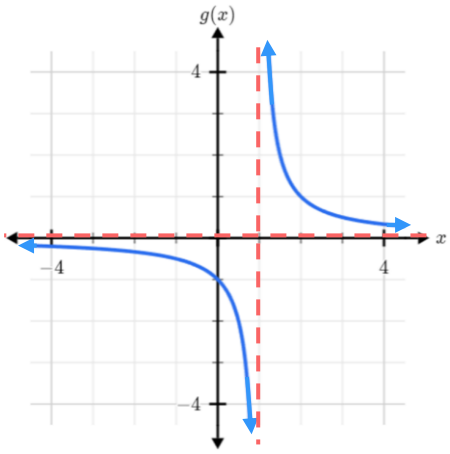
The range is ……



The range is ……



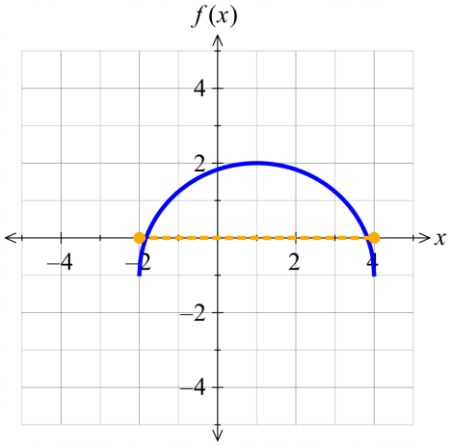
The range is ……



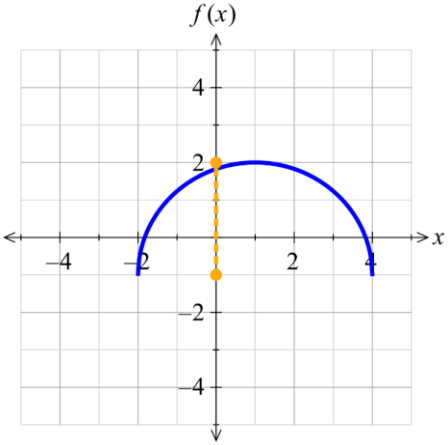
The range is ……

ⓘ Domain and Range of a Function

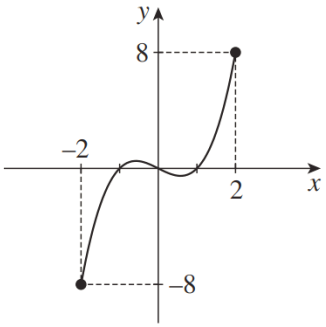
The domain of a function is the set of all allowable x-values.
Visualise it as a shadow on the x-axis.



The range of a function is the set of all possible y-values.
Visualise it as a shadow on the y-axis.

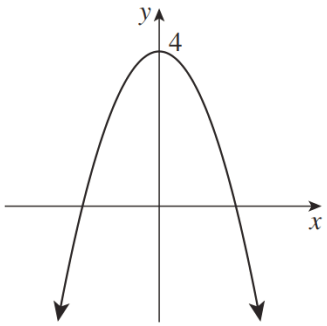


Example Identify domain and range from a graph



Domain: $-2 \leq x \leq 2$ $[-2,2]$

Range: $-8 \leq y \leq 8$ $[-8,8]$

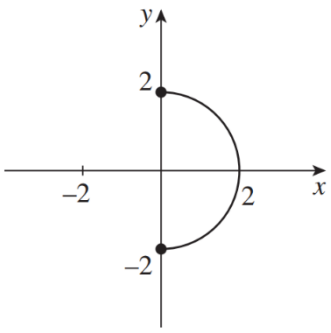


Domain: all real x $x \in \mathbb{R}$

Range: $y \leq 4$ $(-\infty, 4]$

Guided Practice

a



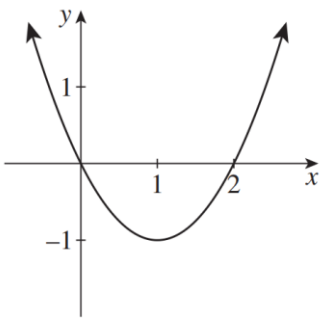
Domain:

$0 \leq x \leq 2$ $[0,2]$

Range:

$-2 \leq y \leq 2$ $[-2,2]$

b



Domain:

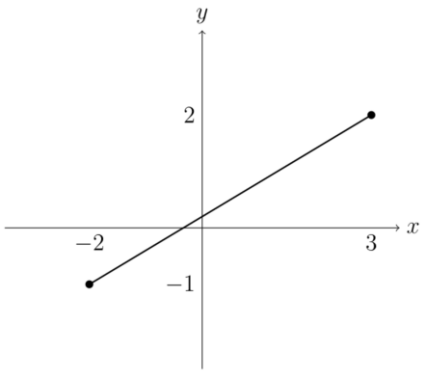
all real x $x \in \mathbb{R}$

Range:

$y \geq -1$ $[-1,\infty)$

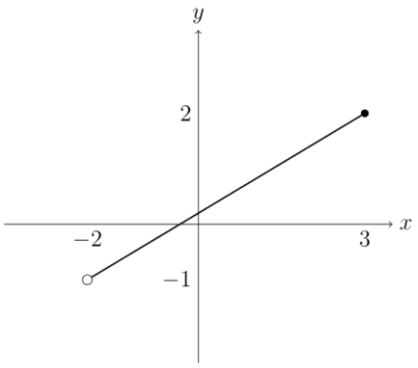
1 Identify the domain and range of these graphs.

a



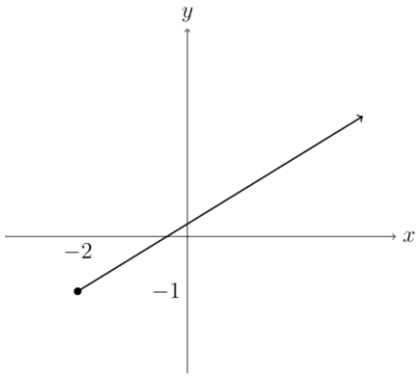
Domain:
Range:

b



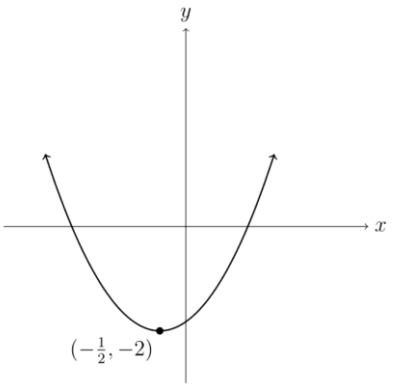
Domain:
Range:

c



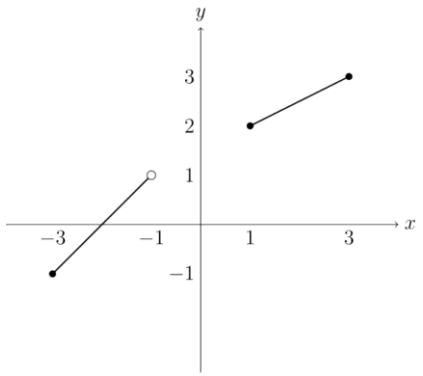
Domain:
Range:

d



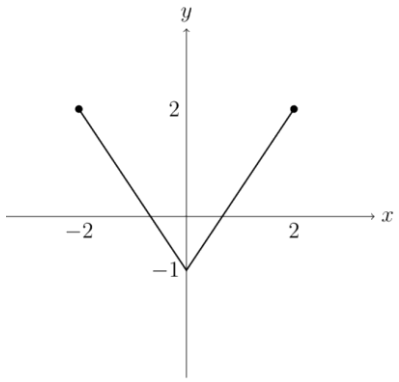
Domain:
Range:

e



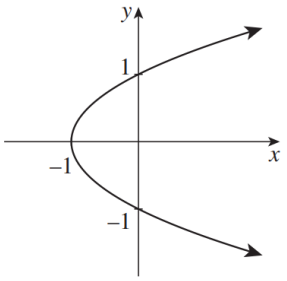
Domain:
Range:

f



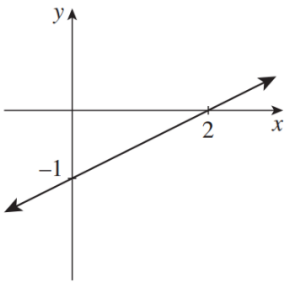
Domain:
Range:

g



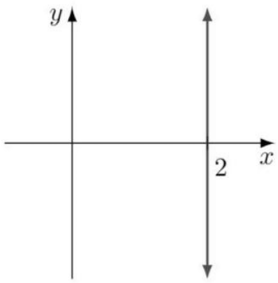
Domain: $[-1, \infty)$
Range: $y \in \mathbb{R}$

h



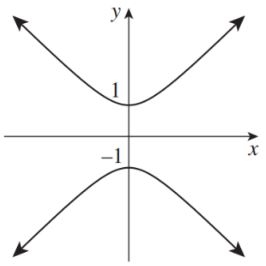
Domain: $x \in \mathbb{R}$
Range: $y \in \mathbb{R}$

i



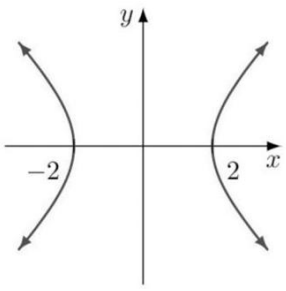
Domain: $x = 2$
Range: $y \in \mathbb{R}$

j



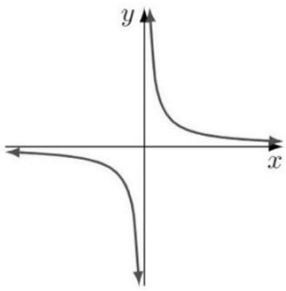
Domain: $x \in \mathbb{R}$
Range: $(-\infty, -1] \cup [1, \infty)$

k



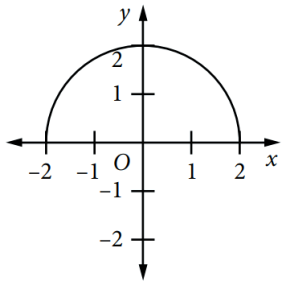
Domain: $(-\infty, -2] \cup [2, \infty)$
Range: $y \in \mathbb{R}$

l



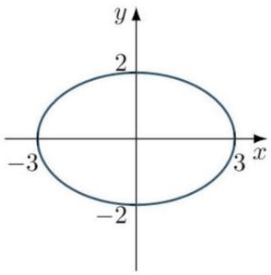
Domain: $(-\infty, 0) \cup (0, \infty)$
Range: $(-\infty, 0) \cup (0, \infty)$

m



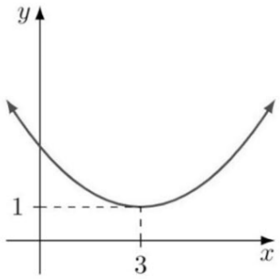
Domain: $[-2,2]$
Range: $[0.,2]$

n



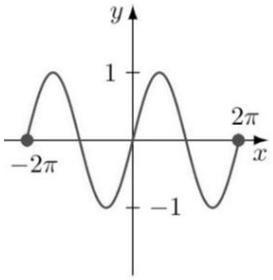
Domain: $[-3,3]$
Range: $[-2.,2]$

o



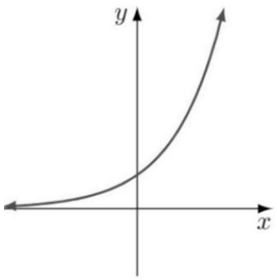
Domain: $x \in \mathbb{R}$
Range: $[1,\infty)$

p



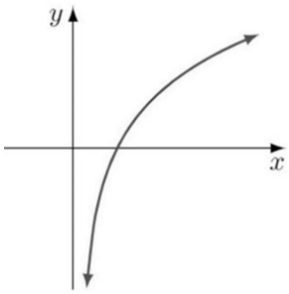
Domain: $[-2\pi, 2\pi]$
Range: $[-1,1]$

q



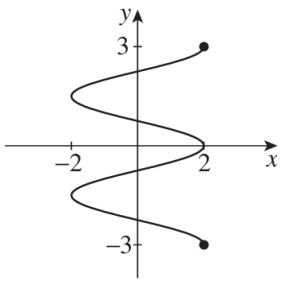
Domain: $x \in \mathbb{R}$
Range: $(0,\infty)$

r



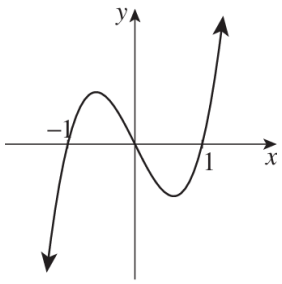
Domain: $(0,\infty)$
Range: $y \in \mathbb{R}$

s



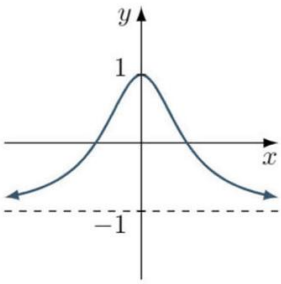
Domain: $[-2,2]$
Range: $[-3,3]$

t



Domain: $x \in \mathbb{R}$
Range: $y \in \mathbb{R}$

u



Domain: $x \in \mathbb{R}$
Range: $(-1,1]$

Natural Domain and Range

ⓘ Natural Domain

A function's **natural domain** includes all possible x-values that produce valid y-values.

Many functions have no restriction on their domain, but some do.

When finding the natural domain, consider that

- **Zero denominators are undefined**.
- **Numbers in square roots must be non-negative**.

Example Find the natural domain of a function

Find the natural domain of:

$$y = \frac{1}{x-2}$$

$x - 2 \neq 0$

$x \neq 2$

Domain: all real numbers except $x = 2$

$(-\infty, 2) \cup (2, \infty)$

Find the natural domain of:

$$y = \sqrt{x-1}$$

$x - 1 \geq 0$

$x \geq 1$

Domain: $x \geq 1$ $[1, \infty)$

Guided Practice

Find the natural domain of each function.

a $y = x^2$

all real x $x \in \mathbb{R}$

b $f(x) = \sqrt{x-5}$

$x \geq 5$ $[5, \infty)$

c $f(x) = \frac{1}{3x}$

all real x except $x = 0$ $(-\infty, 0) \cup (0, \infty)$

d $y = \frac{1}{\sqrt{3-x}}$

$x < 3$ $(-\infty, 3)$

ⓘ Natural Range

The natural range is most easily found from sketching the graph.

To find it algebraically, consider that:

$$\sqrt{\text{expression}} \geq 0, \qquad (\text{expression})^2 \geq 0, \qquad \frac{1}{\text{expression}} \neq 0$$

1 Identify the natural domain of these functions.

a $f(x) = \sqrt{x}$

$x \geq 0$

b $f(x) = \sqrt{x-2}$

$x \geq 2$

c $f(x) = \sqrt{x+2}$

$x \geq -2$

d $f(x) = \sqrt{5-x}$

$x \leq 5$

e $f(x) = \sqrt{7-2x}$

$x \leq \frac{7}{2}$

f $f(x) = \sqrt{2+5x} + 1$

$x \geq -\frac{2}{5}$

2 Identify the natural domain of these functions.

a $f(x) = \frac{1}{x}$

$x \neq 0$

b $f(x) = \frac{1}{x-2}$

$x \neq 2$

c $f(x) = \frac{1}{x+3}$

$x \neq -3$

d $f(x) = \frac{1}{2x-3}$

$x \neq \frac{3}{2}$

e $f(x) = \frac{x}{x+2}$

$x \neq -2$

f $f(x) = \frac{2}{x-7} + 3$

$x \neq 7$

3 Recall that the range of $f(x) = x^2$ is $y \geq 0$ because x^2 cannot be negative.
Use this, or your knowledge of the graph, to determine the range of:

a $f(x) = 4x^2$

$y \geq 0$

b $f(x) = x^2 + 3$

$y \geq 3$

c $f(x) = 3x^2 - 4$

$y \geq -4$

d $f(x) = -x^2$

$y \leq 0$

e $f(x) = -x^2 + 6$

$y \leq 6$

f $f(x) = 5 - x^2$

$y \leq 5$

g $f(x) = 3 - 4x^2$

$y \leq 3$

h $f(x) = (x-3)^2 + 2$

$y \geq 2$

i $f(x) = 10 - (x+2)^2$

$y \leq 10$

4 Find the natural domain of these functions.

a $f(x) = 4x$

$x \in \mathbb{R}$

b $f(x) = \frac{1}{4x}$

$x \neq 0$

c $f(x) = \frac{1}{x+4}$

$x \neq -4$

d $f(x) = \frac{1}{-x+4}$

$x \neq 4$

e $f(x) = \frac{1}{x-4}$

$x \neq 4$

f $f(x) = \frac{1}{6x-4}$

$x \neq \frac{2}{3}$

g $f(x) = \frac{3}{6x-4}$

$x \neq \frac{2}{3}$

h $f(x) = -\frac{3}{6x-4}$

i $f(x) = 3x$

$x \in \mathbb{R}$

j $f(x) = \sqrt{3x}$

$x \geq 0$

k $f(x) = \sqrt{x+3}$

$x \geq -3$

l $f(x) = \sqrt{-x+3}$

$x \leq 3$

m $f(x) = \sqrt{x-3}$

$x \geq 3$

n $f(x) = \sqrt{8x-3}$

$x \geq \frac{3}{8}$

o $f(x) = 5\sqrt{8x-3}$

$x \geq \frac{3}{8}$

p $f(x) = -5\sqrt{8x-3}$

q $f(x) = x$

$x \in \mathbb{R}$

r $f(x) = \frac{1}{\sqrt{x}}$

> 0

s $f(x) = \frac{1}{\sqrt{5x}}$

$x > 0$

t $f(x) = \frac{1}{\sqrt{5x+1}}$

$x > -\frac{1}{5}$

u $f(x) = \frac{2}{\sqrt{5x+1}}$

$x > -\frac{1}{5}$

v $f(x) = \frac{2}{\sqrt{5x-1}}$

$x > \frac{1}{5}$

w $f(x) = \frac{2}{\sqrt{1-5x}}$

$x < \frac{1}{5}$

x $f(x) = -\frac{2}{\sqrt{1-5x}}$

5 Recall that the range of $f(x) = \sqrt{x}$ is $y \geq 0$ because $\sqrt{x}$ cannot be negative.
Use this to determine the range of:

a $f(x) = 2\sqrt{x}$

$y \geq 0$

b $f(x) = \sqrt{x} + 3$

$y \geq 3$

c $f(x) = \sqrt{x} - 5$

$y \geq -5$

d $f(x) = -\sqrt{x}$

$y \leq 0$

e $f(x) = -\sqrt{x} + 4$

$y \leq 4$

f $f(x) = 3 - \sqrt{x}$

$y \leq 3$

6 Identify the natural domain of these functions.

a $f(x) = \frac{2}{\sqrt{1-x}}$

$x < 1$

b $f(x) = \frac{1}{\sqrt{2x-3}}$

$x > \frac{3}{2}$

c $f(x) = \frac{x}{\sqrt{x+2}}$

$x > -2$

d $f(x) = \frac{1}{x^2-4}$

$x \neq 2$ and $x \neq -2$

e $f(x) = \frac{1}{x^2+x}$

$x \neq 0$ and $x \neq -1$

f $f(x) = \frac{1}{x^2-5x+6}$

$x \neq 2$ and $x \neq 3$

7 Using graphing software, or otherwise, sketch $f(x) = (x-2)^2 + 1$.

a State the range of $f(x)$.

$y \geq 1$

b Similarly, find the range of:

i $f(x) = (x+1)^2 - 4$

$y \geq -4$

ii $f(x) = (x-2)^2 + 3$

$y \geq 3$

iii $f(x) = (x-2)^2 - 5$

$y \geq -5$

iv $f(x) = x^2 + 6x + 11$

$y \geq 2$

c What do you notice about the range of the function $f(x) = (x-a)^2 + k$? Why is this?

The range is $y \geq k$

This is because the coordinate $y = k$ is part of the minimum turning point of the concave up parabola

8 State the range, start with $\sqrt{\text{expression}} \geq 0$.

a $f(x) = \sqrt{x}$

$y \geq 0$

b $f(x) = \sqrt{x} + 3$

$y \geq 3$

c $f(x) = \sqrt{x-4} + 3$

$y \geq 3$

d $f(x) = 2\sqrt{x-4} + 3$

$y \geq 3$

e $f(x) = 5 - \sqrt{x-4}$

$y \leq 5$

f $f(x) = 5 - 2\sqrt{x-4}$

$y \leq 5$

9 State the range, start with $x^2 \geq 0$.

a $f(x) = x^2$

$y \geq 0$

b $f(x) = x^2 + 4$

$y \geq 4$

c $f(x) = 3x^2 + 4$

$y \geq 4$

d $f(x) = 4 - 3x^2$

$y \leq 4$

e $f(x) = 4 - 3(x-2)^2$

$y \leq 4$

f $f(x) = -2(x+1)^2 - 3$

$y \leq -3$

10 State the range, start with the fact that a non-zero reciprocal term cannot equal zero $\frac{1}{x} \neq 0$.

a $f(x) = \frac{1}{x}$

$y \neq 0$

b $f(x) = \frac{1}{x} + 2$

$y \neq 2$

c $f(x) = \frac{4}{x} + 2$

$y \neq 2$

d $f(x) = \frac{4}{x} - 3$

$y \neq -3$

e $f(x) = \frac{4}{x-1} - 3$

$y \neq -3$

f $f(x) = 5 - \frac{2}{x+4}$

$y \neq 5$

11 State the range of these functions, using $x^2 \geq 0$, $\sqrt{x} \geq 0$, $\frac{1}{x} \neq 0$

a $f(x) = 7 - x^2$

$y \leq 7$

b $f(x) = \sqrt{x+6} - 4$

$y \geq -4$

c $f(x) = \frac{3}{x-2} + 5$

$y \neq 5$

d $f(x) = 2(x+1)^2 - 6$

$y \geq -6$

e $f(x) = 8 - \sqrt{3x-1}$

$y \leq 8$

f $f(x) = \frac{3}{x-2} - 4$

$y \neq -4$

12 State the natural domain of these functions.

a $f(x) = \frac{\sqrt{x}}{x-2}$

$[0,2) \cup (2, \infty)$

b $f(x) = \frac{\sqrt{x+1}}{x-2}$

$[-1,2) \cup (2, \infty)$

c $f(x) = \sqrt{x+1} + \sqrt{5-x}$

$[-1,5]$

d $f(x) = \sqrt{x+1} + \sqrt{3-x}$

$[-1,3]$

e $f(x) = \frac{1}{\sqrt{x+1}}$

$x > -1$

f $f(x) = \frac{1}{\sqrt{3-x}}$

$x < 3$

g $f(x) = \sqrt{x+1} + \frac{1}{\sqrt{5-x}}$

$[-1,5)$

h $f(x) = \sqrt{x+4} + \frac{1}{x-1}$

$[-4,1) \cup (1, \infty)$

i $f(x) = \frac{\sqrt{x+3}}{x-1}$

$[-3,1) \cup (1, \infty)$

j $f(x) = \frac{\sqrt{x+3}}{x+1}$

$[-3,-1) \cup (-1, \infty)$

k $f(x) = \frac{\sqrt{5-x}}{x+1}$

$(-\infty,-1) \cup (-1,5]$

l $f(x) = \frac{\sqrt{5-x}}{x-6}$

$(-\infty,5]$

13

a Determine the range of $y = 9 - x^2$ and hence find the maximum value of y.

Range $y \leq 9$, so max value: $y = 9$

b Hence, find the range of $f(x) = \sqrt{9 - x^2}$

$0 \leq y \leq 3$

c Use a similar technique to find the range of $y = \sqrt{4 - x^2}$

$0 \leq y \leq 2$

d Hence determine the range of $y = \frac{1}{\sqrt{4-x^2}}$

Since $0 \leq \sqrt{4 - x^2} \leq 2$,

Then $\frac{1}{\sqrt{4-x^2}} \geq \frac{1}{2}$

(need to reverse inequality when reciprocating)

e By considering the minimum value, find the range of $y = \sqrt{9 + x^2}$

Minimum value of $9 + x^2$ is 9.

$y \geq 3$

f Hence find the range of $y = \frac{1}{\sqrt{9+x^2}}$

Consider that $3 \leq \sqrt{9 + x^2} < \infty$,

then $0 < \frac{1}{\sqrt{9 + x^2}} \leq \frac{1}{3}$

(need to reverse inequality when reciprocating)